LEFTMOST REASSOCIATION OF DYCK COMPONENTS: EXACT TRANSIENT LAYERS AND TERMINAL DEPTH FIBRES

ANONYMOUS

ABSTRACT. Let a nonempty Dyck path have primitive factorisation $P = C_1 \cdots C_k$, with $C_1 = UAD$. We iterate the deterministic rule that fixes P when $k = 1$ and otherwise replaces its first two components by UAC_2D. The complete orbit is explicit: after t updates, the first component has absorbed $C_2, \dots, C_{t+1}$. Consequently the entry time is exactly $k-1$, all recurrent paths are fixed and primitive, and the unique path of maximum depth $n-1$ in semilength n is $(UD)^n$. The number of paths at depth $k-1$ is

$$\frac{k}{2n-k}\binom{2n-k}{n}.$$

We also resolve every terminal basin by depth. If a fixed target is UQD and the interior Q has r primitive factors, then it has exactly one basin state at each depth $0, \dots, r$. Hence its depth-fibre polynomial is $1 + u + \cdots + u^r$, and the unique largest terminal fibre has size n. First-return decomposition, Catalan and ballot enumeration, the atomic Tamari cover, and generic coefficient extraction are treated as background inputs. This is an owner-thin anonymous draft, and external circulation remains on hold.

1. Introduction

A local cover relation does not by itself determine a dynamical system: one must also specify which available cover is chosen at each state. This note fixes a particularly rigid selector on Dyck paths. At every nonfixed state, we perform the reassociation at the leftmost return to ground level. The selected down-step crosses the next primitive Dyck component, and no other part of the path changes.

The atomic reassociation belongs to the Tamari setting. Associativity orders and their lattice property go back to Huang and Tamari [2]; path formulations of Tamari-type covers also appear in the m-Tamari literature [1]. Primitive decomposition of Dyck words and enumeration by the number of primitive components are likewise established topics [3], while Catalan and ballot enumeration are standard background [4]. We assign all of these ingredients zero contribution credit.

The purpose here is narrower: to record the exact finite dynamics induced by the repeated-leftmost selector. The argument has four parts.

(1) We give a closed formula for every iterate. It turns the number of primitive factors into the pointwise clock $\tau(P) = \kappa(P) - 1$ and identifies $(UD)^n$ as the unique depth-$n-1$ path.
(2) We derive the complete temporal census: depth $k-1$ contains $\frac{k}{2n-k}\binom{2n-k}{n}$ states.
(3) For every fixed target and every feasible depth, we construct one basin state and prove that no second state exists at that depth.
(4) We derive the depth-fibre polynomial and classify the unique target of largest fibre.

The closed iterate in lemma 3.1 is the common mechanism: it proves the clock and leaves only a suffix cut to invert an endpoint. No figure is needed because that formula exposes the whole state change.

This package is owner-thin. The source boundary in section 7 is a conservative credit assignment, not a priority or originality decision. The draft is maintained for internal theorem evaluation under HOLD__EXTERNAL.

2. Dyck factors and the literal map

A *Dyck path of semilength* n is a word in U, D with n occurrences of each letter such that every prefix contains at least as many U's as D's. Let $\mathcal{D}_n$ denote the set of such paths. A nonempty Dyck path is *primitive* if its only return to height zero occurs after its final step. Cutting at all positive returns gives a unique factorisation

$$P = C_1 C_2 \cdots C_k \tag{1}$$

into primitive Dyck paths. We write $\kappa(P) = k$.

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Every primitive first factor has a unique first-return form $C_1 = UAD$ with A a possibly empty Dyck path. Define

$$\Phi_n(P) = \begin{cases} P, & k = 1, \\ UAC_2D\,C_3 \cdots C_k, & k \geq 2. \end{cases} \tag{2}$$

We suppress the subscript n when the semilength is fixed. The word AC_2 is a Dyck path, so UAC_2D is primitive. Thus (2) lies in $\mathcal{D}_n$ and is well defined.

Let $\tau(P)$ be the least $t \geq 0$ for which $\Phi^t(P)$ is fixed, and put

$$\mathrm{E}(P) = \Phi^{\tau(P)}(P).$$

A path is *recurrent* when it lies on a directed cycle of Φ_n. For a fixed target T, define its depth-fibre polynomial

$$\mathcal{B}_T(u) = \sum_{P:\,\mathrm{E}(P)=T} u^{\tau(P)}. \tag{3}$$

The complete claim package follows. We use $\mathrm{Cat}_m = \frac{1}{m+1}\binom{2m}{m}$.

Theorem 2.1 (Clock, layers, and terminal depth fibres)**.** *For every $n \geq 1$, the map Φ_n has the following properties.*

(i) *Every nonfixed update reduces κ by exactly one. The recurrent paths are exactly the fixed paths, these are exactly the primitive paths, and there are Cat_{n-1} of them.*

(ii) *If P has k primitive factors, then*

$$\tau(P) = k - 1. \tag{4}$$

The maximum entry time is $n - 1$, attained uniquely by $(UD)^n$.

(iii) *For $1 \leq k \leq n$, the number of paths at depth $k - 1$ is*

$$\bigl|\{P \in \mathcal{D}_n : \tau(P) = k - 1\}\bigr| = \frac{k}{2n-k}\binom{2n-k}{n}. \tag{5}$$

(iv) *Let $T = UQD$ be fixed, and factor its interior as $Q = Q_1 \cdots Q_r$ into primitive paths. For every $0 \leq d \leq r$, the unique basin state of T at depth d is*

$$P_d = \bigl(UQ_1 \cdots Q_{r-d}D\bigr)Q_{r-d+1} \cdots Q_r, \tag{6}$$

with the empty-prefix and empty-suffix conventions. Consequently

$$\mathcal{B}_T(u) = 1 + u + \cdots + u^r. \tag{7}$$

The unique terminal fibre of maximum size is the fibre over

$$T_n^{\max} = U(UD)^{n-1}D, \tag{8}$$

and it contains n states.

3. The factor-count clock

The first primitive component absorbs exactly one following component per round. The next lemma makes that statement precise at every time.

Lemma 3.1 (Closed orbit)**.** *Let $P = C_1 \cdots C_k$, where $C_1 = UAD$. For every $0 \leq t \leq k - 1$,*

$$\Phi^t(P) = UA\,C_2 \cdots C_{t+1}D\,C_{t+2} \cdots C_k. \tag{9}$$

When $t = 0$, the block $C_2 \cdots C_{t+1}$ is empty; when $t = k - 1$, the suffix $C_{t+2} \cdots C_k$ is empty.

Proof. At $t = 0$, (9) reads $UADC_2 \cdots C_k = P$. Suppose the formula holds for some $t < k - 1$. The word

$$UA\,C_2 \cdots C_{t+1}D$$

is primitive: after its initial U, the Dyck word $AC_2 \cdots C_{t+1}$ stays at or above height one, and the displayed D is its first return to height zero. Hence the primitive factorisation of the right-hand side of (9) consists of that first word, followed by $C_{t+2}, \ldots, C_k$. Applying (2) moves the closing D across C_{t+2} and gives

$$UA\,C_2 \cdots C_{t+1}C_{t+2}D\,C_{t+3} \cdots C_k,$$

which is (9) at time $t + 1$. Induction proves the claim. □

Corollary 3.2 (Recurrent set and pointwise clock)**.** *The statements in theorem 2.1(i)–(ii) hold.*

Proof. For $t < k-1$, the factorisation displayed in the proof of lemma 3.1 has one first component and $k-t-1$ untouched components, for a total of $k-t$ primitive factors. Thus each nonfixed step decreases κ by one. At $t = k-1$, the formula is the single primitive path

$$\mathrm{E}(P) = UAC_2 \cdots C_k D. \tag{10}$$

It is fixed, so $\tau(P) = k-1$.

If $k = 1$, the definition fixes P. If $k > 1$, the strict decrease of κ prevents P from lying on a cycle. Fixed and recurrent paths are therefore exactly the primitive paths. Deleting the first U and last D is a bijection from primitive paths in $\mathcal{D}_n$ to $\mathcal{D}_{n-1}$, giving Cat_{n-1} fixed paths.

Every primitive factor has semilength at least one, so $k \le n$. Equality holds only when all k factors have semilength one, and the only such factor is UD. In view of (4), the depth is at most $n-1$, with equality exactly at $(UD)^n$. □

The proof gives more than a global bound: the component statistic is the exact amount of time remaining at every state. In particular, no transient can merge with another transient of a different factor count before their depths agree.

4. Complete temporal layers

The clock reduces temporal enumeration to a component census. We include the coefficient calculation to fix the boundary case and normalisation, while treating the enumerative machinery as background.

Let

$$C(z) = \sum_{m \ge 0} \mathrm{Cat}_m \, z^m$$

be the Catalan series, which satisfies $C(z) = 1 + zC(z)^2$. A primitive path is UAD with A an arbitrary Dyck path, so the semilength generating function for one primitive component is

$$R(z) = zC(z). \tag{11}$$

Proposition 4.1 (Ballot layer formula)**.** *For $1 \le k \le n$, the number of paths in $\mathcal{D}_n$ with exactly k primitive factors is the right-hand side of* (5)*.*

Proof. A path with k primitive factors is an ordered sequence of k primitive paths. By (11), its number is

$$[z^n]R(z)^k = [z^{n-k}]C(z)^k. \tag{12}$$

Put $m = n-k$ and $W = C - 1$. Then $W = z(1+W)^2$. If $m = 0$, the coefficient in (12) is 1, which agrees with $k(2n-k)^{-1}\binom{2n-k}{n} = 1$ because $n = k$.

For $m \ge 1$, Lagrange inversion gives

$$\begin{aligned}
[z^m]C(z)^k &= [z^m](1+W)^k \\
&= \frac{1}{m}[w^{m-1}]k(1+w)^{k-1}(1+w)^{2m} \\
&= \frac{k}{m}\binom{2m+k-1}{m-1} \\
&= \frac{k}{2m+k}\binom{2m+k}{m}.
\end{aligned}$$

Substituting $m = n-k$ and using binomial symmetry yields

$$\frac{k}{2n-k}\binom{2n-k}{n-k} = \frac{k}{2n-k}\binom{2n-k}{n},$$

as required. □

Combining corollary 3.2 and proposition 4.1 proves theorem 2.1(iii). It also gives the complete temporal polynomial

$$\sum_{P \in \mathcal{D}_n} u^{\tau(P)} = \sum_{k=1}^{n} \frac{k}{2n-k}\binom{2n-k}{n} u^{k-1}. \tag{13}$$

Across all semilengths, the same sequence construction yields

$$\sum_{n \ge 1} \sum_{P \in \mathcal{D}_n} u^{\tau(P)} z^n = \frac{R(z)}{1-uR(z)} = \frac{zC(z)}{1-uzC(z)}. \tag{14}$$

Equations (13)–(14) package the clock after it has been proved; generic generating-function extraction is not part of the residual theorem claim.

5. The terminal depth-fibre atlas

The endpoint formula (10) absorbs every component after the first into the interior of one primitive path. Inverting it amounts to choosing how many final primitive components of the target interior should be moved back outside.

Theorem 5.1 (One source at every feasible depth). *Let $T = UQD \in \mathcal{D}_n$ be primitive, and let $Q = Q_1 \cdots Q_r$ be the primitive factorisation of its interior. For each $0 \leq d \leq r$, the path P_d in (6) is the unique path satisfying*

$$\mathrm{E}(P_d) = T, \qquad \tau(P_d) = d.$$

There is no basin state of T at any other depth.

Proof. Fix $d \in \{0, \ldots, r\}$. The word

$$UQ_1 \cdots Q_{r-d}D$$

is primitive because its interior is a Dyck path. The suffix $Q_{r-d+1}, \ldots, Q_r$ consists of d primitive paths. Hence P_d has exactly $d+1$ primitive factors. By corollary 3.2, its depth is d, and the endpoint formula gives

$$\mathrm{E}(P_d) = UQ_1 \cdots Q_{r-d}Q_{r-d+1} \cdots Q_rD = T.$$

This proves existence.

For uniqueness, take any P with $\mathrm{E}(P) = T$ and $\tau(P) = d$. The clock implies that P has $d+1$ primitive factors, say

$$P = B_1B_2 \cdots B_{d+1}, \qquad B_1 = UAD.$$

Its endpoint is $UAB_2 \cdots B_{d+1}D$. Equality with UQD implies

$$Q = AB_2 \cdots B_{d+1}. \tag{15}$$

Because A is a Dyck path, the boundary immediately after A is a return of Q unless A is empty. Each B_i for $i \geq 2$ is primitive. Therefore the unique factorisation of Q into primitive paths forces

$$A = Q_1 \cdots Q_{r-d}, \qquad (B_2, \ldots, B_{d+1}) = (Q_{r-d+1}, \ldots, Q_r).$$

Thus $P = P_d$. The same argument shows that d must lie between 0 and r, so no other depth occurs. ☐

The boundary cases are included in the construction. If $r = 0$, then $T = UD$ and only $d = 0$ occurs. For arbitrary r, the case $d = 0$ gives $P_0 = T$, while $d = r$ gives $P_r = (UD)Q_1 \cdots Q_r$.

Corollary 5.2 (Fibre polynomial and unique maximum). *Equations (7)–(8) hold.*

Proof. By theorem 5.1, the coefficient of u^d in $\mathcal{B}_T(u)$ is one for $0 \leq d \leq r$ and zero otherwise. This proves (7); in particular, the ordinary fibre size is $\mathcal{B}_T(1) = r + 1$.

The interior Q has semilength $n - 1$. Since every Q_i has positive semilength, $r \leq n - 1$, and hence $\mathcal{B}_T(1) \leq n$. Equality forces all $r = n - 1$ factors to have semilength one. The only primitive factor of semilength one is UD, so equality occurs exactly when $Q = (UD)^{n-1}$. This gives the unique target (8). ☐

The terminal atlas is pointwise: it identifies the source word, not merely the number of sources. It also records when each source enters the fixed set, rather than collapsing all depths into one ordinary fibre count.

6. Exact finite controls

The standard-library program `verify_p144.py` supplies finite counterexample pressure. It generates every Dyck path of semilength at most twelve. The literal map is implemented from the first two return positions, whereas the closed iterate is implemented separately from the initial primitive-factor list. Basin profiles are accumulated from the enumerated functional graph and compared with the constructive source formula for every fixed target and every feasible depth.

Table 1. Selected exact control sizes. The audit is exhaustive within each displayed semilength and uses no random sampling.

n	states	fixed targets	maximum depth	maximum fibre	assertions
1	1	1	0	1	24
4	14	5	3	4	277
8	1,430	429	7	8	28,901
12	208,012	58,786	11	12	4,308,839

Across semilengths 1 through 12, the run covers 290,511 states and 82,500 fixed targets. It performs 6,005,502 exact assertions and ends in `STATUS=PASS`. The frozen transcript is `verification_output.txt`. The audit checks closure, factor drop, every iterate, every temporal layer, the unique sharp source, all terminal depth profiles, and the unique maximum target.

Finite enumeration does not prove an all-parameter identity. The proofs in sections 3 to 5 establish the theorem; the program can only reveal errors within its exhaustive range. It also supplies no evidence about priority or ownership.

7. Ownership boundary and conclusion

The carrier and several ingredients have direct owners. Huang and Tamari establish the semi-associative order and lattice background [2]. In a path model, the local operation in (2) is the familiar Tamari-type exchange of a down-step with the following excursion; path cover formulations are part of the m-Tamari setting [1]. Accordingly, the atomic update receives zero contribution credit.

Prime factorisation is equally explicit background. Panayotopoulos and Sapounakis study decomposition of Dyck words into prime subwords and evaluate the number with a prescribed component count [3]. Thus the primitive-factor language and the ballot layer numbers in (5) are not residual claims. Catalan enumeration, first-return arguments, ballot methods, and the generating-function calculations are also treated as standard [4].

After these deductions, the internal theorem package consists of the fixed repeated-leftmost scheduler, its closed orbit and exact pointwise clock, and the targetwise statement that every feasible depth has one explicitly identified source. The same mechanism yields a complete transient census and the unique largest terminal fibre, but those conclusions are presented with their background enumeration fully credited.

The scope is deliberately limited. We do not classify alternative Tamari schedulers or compare arbitrary maximal chains, and this bounded source record does not settle ownership of the conjunction proved here. The manuscript is therefore owner-thin and remains an anonymous internal draft under HOLD_EXTERNAL.

References

[1] Mireille Bousquet-Mélou, Éric Fusy, and Louis-François Préville-Ratelle. The number of intervals in the m-Tamari lattices. *The Electronic Journal of Combinatorics*, 18(2):P31, 2012. doi: 10.37236/2027. URL https://doi.org/10.37236/2027.

[2] Samuel Huang and Dov Tamari. Problems of associativity: A simple proof for the lattice property of systems ordered by a semi-associative law. *Journal of Combinatorial Theory, Series A*, 13(1):7–13, 1972. doi: 10.1016/0097-3165(72)90003-9. URL https://doi.org/10.1016/0097-3165(72)90003-9.

[3] A. Panayotopoulos and A. Sapounakis. On the prime decomposition of Dyck words. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 40:33–39, 2002.

[4] Richard P. Stanley. *Catalan Numbers*. Cambridge University Press, 2015. ISBN 978-1-107-07509-2. doi: 10.1017/CBO9781139871495. URL https://doi.org/10.1017/CBO9781139871495.